# MATH 127 Calculus for the Sciences

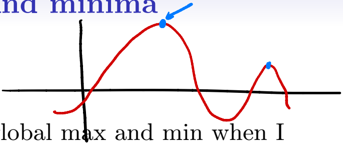
Lecture 29

# Today's lecture

**Last time**
Curve sketching

**This time**
**Course note coverage** Section 4.1.1
Absolute maxima/minima
a.k.a Global maxima and minima

# Absolute maxima and minima



Didn't we do this already? I talked about global max and min when I talked about local max and min. What else do you want me to say?

**Definition**

- An input $x = a$ is a **global maximum** of $f$ if

$$f(a) \geq f(x) \text{ for all } x \text{ in the domain of } f(x).$$

- An input $x = a$ is a **global minimum** of $f$ if

$$f(a) \leq f(x) \text{ for all } x \text{ in the domain of } f(x).$$

Well, turns out there's some theory you can learn in terms of locating the maxima/minima.

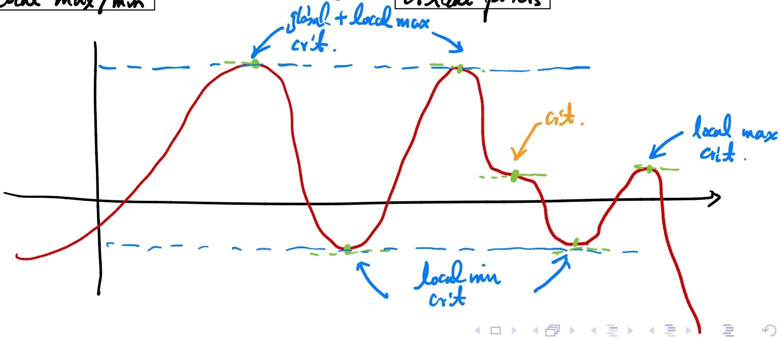
## Absolute maxima and minima

Fermat's theorem says that a **local max/min** must be a $\boxed{\text{critical point}}$.

This means to find the **local max/min**, we only need to look for them among $\boxed{\text{critical points}}$.

I say that a **global max/min** must be a $\boxed{\text{local max/min}}$.

So to find global max/min, we only need to look for them among $\boxed{\text{local max/min}}$, which lie among the $\boxed{\text{critical points}}$.

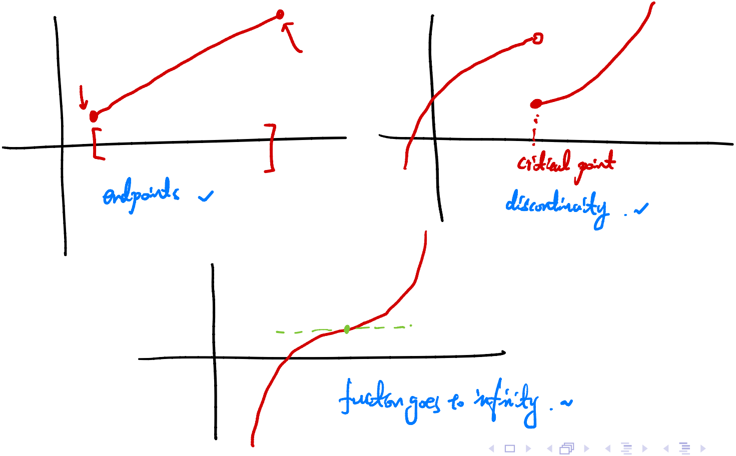


# Absolute maxima and minima

**Proposed method**

We can list out all critical points, then the biggest one is the global max and the smallest one is the global min.

But this method does not work in general, what are the flaws?

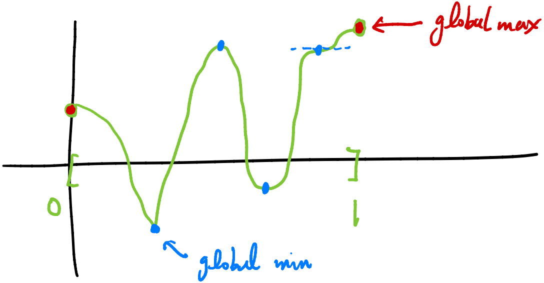


# Absolute maxima and minima

**Correct method**

Suppose we know $f(x)$ is **continuous** on a **closed interval** $[a, b]$.
We can list out critical points and the edge points, then the biggest one is the global max and the smallest one is the global min on the interval $[a, b]$.

**Example** If I have a function continuous on the interval $[0, 1]$, would the flaws we saw previously still occur?



# Extreme value theorem

**Theorem** *If $f$ is* ***continuous*** *on a* ***closed interval*** $[a, b]$*, then $f$ must have a global* *maximum* *with respect to* $[a, b]$

**Remark** So this says $f$ must have a global max, but what about a global min?

Let $g = -f$. Then $g$ is continuous by continuity rule
and defined on closed interval $[a,b]$.

By extreme value theorem, there exists a global max of $g$.
so for some $c$ in $[a,b]$

$g(c) \geq g(x)$ for every other $x$ in $[a,b]$

$-f(c) \geq -f(x)$

$f(c) \leq f(x)$. So $c$ is a global minimum of $f$.

# Closed interval method

Suppose $f(x)$ is a $\boxed{\text{continuous}}$ function defined on a $\boxed{\text{closed interval } [a, b]}$. Then you can do the following to find its global max and min:

1. Find all critical points.
2. List out the output at the two edge points of $[a, b]$ and at those critical points.
3. The biggest ones are global max, and the smallest ones are global min.

**Warning:**

- This only works if $f$ is **contunious** and defined on a **closed interval**
- There might be multiple points where $f$ achieves global max/min.

# Closed interval method

**Example** Find the global max and global min, including at which inputs they are achieved, of the function $f(x) = \sin\left(\frac{1}{x}\right)$ on the closed interval $\left[\frac{1}{100\pi}, \frac{2}{\pi}\right]$.

Step 1: - $\frac{1}{x}$ is continuous on $\left[\frac{1}{100\pi}, \frac{2}{\pi}\right]$, so $\sin(\frac{1}{x})$ is also continuous by continuity rule.
- The domain $\left[\frac{1}{100\pi}, \frac{2}{\pi}\right]$ is closed.

Step 2: Endpoints: $f(\frac{1}{100\pi}) = \sin(100\pi) = 0$. $f(\frac{2}{\pi}) = \sin(\frac{\pi}{2}) = 1$.

Step 3: Critical points: $f'(x) = -\frac{1}{x^2}\cos(\frac{1}{x})$.

$f'(x) = 0$ when $\cos(u) = 0$. $u = \frac{1}{x}$

$\frac{2}{201\pi} < \frac{2}{200\pi} = \frac{1}{100\pi}$

So $u = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$-\frac{\pi}{2}, -\frac{3\pi}{2}, \ldots$

$x = \frac{2}{\pi}, \frac{2}{3\pi}, \ldots, \frac{2}{199\pi}, \frac{2}{201\pi}$ not in domain

$-\frac{2}{\pi}, -\frac{2}{3\pi}, \ldots$ not in domain

# Closed interval method

**Example** Find the global max and global min, including at which inputs they are achieved, of the function $f(x) = \sin\left(\frac{1}{x}\right)$ on the closed interval $\left[\frac{1}{100\pi}, \frac{2}{\pi}\right]$.

$f(\frac{2}{\pi}) = \sin(\frac{\pi}{2}) = 1,\quad f(\frac{2}{3\pi}) = \sin(\frac{3\pi}{2}) = -1$

$f(\frac{2}{5\pi}) = \sin(\frac{5\pi}{2}) = 1,\quad f(\frac{2}{7\pi}) = \sin[\frac{7\pi}{2}) = -1$

$\vdots$

$f(\frac{2}{197\pi}) = \sin[\frac{197\pi}{2}) = 1,\quad f(\frac{2}{199\pi}) = \sin(\frac{199\pi}{2}) = -1.$

Step 4 : Conclusion:

Global max: $\frac{2}{\pi}, \frac{2}{5\pi}, \ldots, \frac{2}{197\pi}$.

Global min: $\frac{2}{3\pi}, \frac{2}{7\pi}, \ldots, \frac{2}{199\pi}$.